

Distributed Systems - Course Introduction

Duration: 10 minutes.

* Indicates required question

1. Email *

2. Full name *

3. Student ID *

4. I want to be a technical architect. Should I take this course? *

1 point

Mark only one oval.

☐ Yes

☐ No

5. I want to open a start-up company or work freelance or get higher salary.
Should I take this course?

* 1 point

Mark only one oval.

☐ Yes

☐ No

6. I do NOT like programming. Should I take this course? *

1 point

Mark only one oval.

☐ Yes

☐ No

7. What will I learn from this course? *

2 points

Check all that apply.

☐ Software Architecture

☐ Microservices.

☐ Tools and Frameworks for Distributed Systems: The Go Programming Language, Modules and RPC.

☐ Replication: State Machine Replication, Primary-Backup Replication, Failure Detectors, Bully Algorithm, Ring Election, Multi-Leader Replication, Leaderless Replication, Cristian's Algorithm, Berkeley Algorithm, Lamport Clocks, Vector Clocks, Version Vectors, Chain Replication.

☐ Consensus: Raft, Paxos, Viewstamped Replication, Zab.

☐ Sharding: Hash Function, Luhn Algorithm, Consistent Hashing, Gossip Protocol, Merkle Trees.

☐ Distributed Transactions: ACID, Two-Phase Commit, Three-Phase Commit, Serializability, Two-Phase Locking, Distributed Locks, Chubby, ZooKeeper.

☐ Consistency Models: Linearizability, Spanner, COPS, Bayou, Dynamo, Cassandra.

☐ Distributed Architecture and Programming: PyTorch, Tensorflow, Spark, Kubernetes, Bitcoin, Smart Contracts, Streaming Video Engine (SVE), Akamai.

☐ Impossibilities: CAP, PRAM, SNOW, PORT.

☐ Amazon Web Services, Google Cloud, Microsoft Azure.

8. How will I be evaluated? *

2 points

Mark only one oval.

- ☐ 2 individual tests (20%), 1 team project (40%), and 1 final oral exam (40%)
- ☐ 2 individual labs (20%), 2 team presentations (20%), 1 team project (20%), and 1 final oral exam (40%)
- ☐ 1 mid-term written exam (20%) and 1 final written exam (80%)
- ☐ 6 individual labs (60%) and 1 final oral exam (40%)

9. Can I get access to the lecture slide PDFs? *

1 point

Mark only one oval.

- ☐ No, you cannot. You have to attend lectures and take notes.
- ☐ Yes, you can get them from Moodle when the course is about to end. However, you should also attend lectures and take notes to get core knowledge with specific examples that can help you reduce your reading time significantly.

10. Do I have to attend lectures? *

2 points

Mark only one oval.

- ☐ Yes, you do. Otherwise you will fail the course.
- ☐ No, you do not. However, you need to complete all the individual labs in class if you want to get higher score. You can also get bonus points in the lectures.

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